



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal to Ramnath Agencies Nangavalli

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

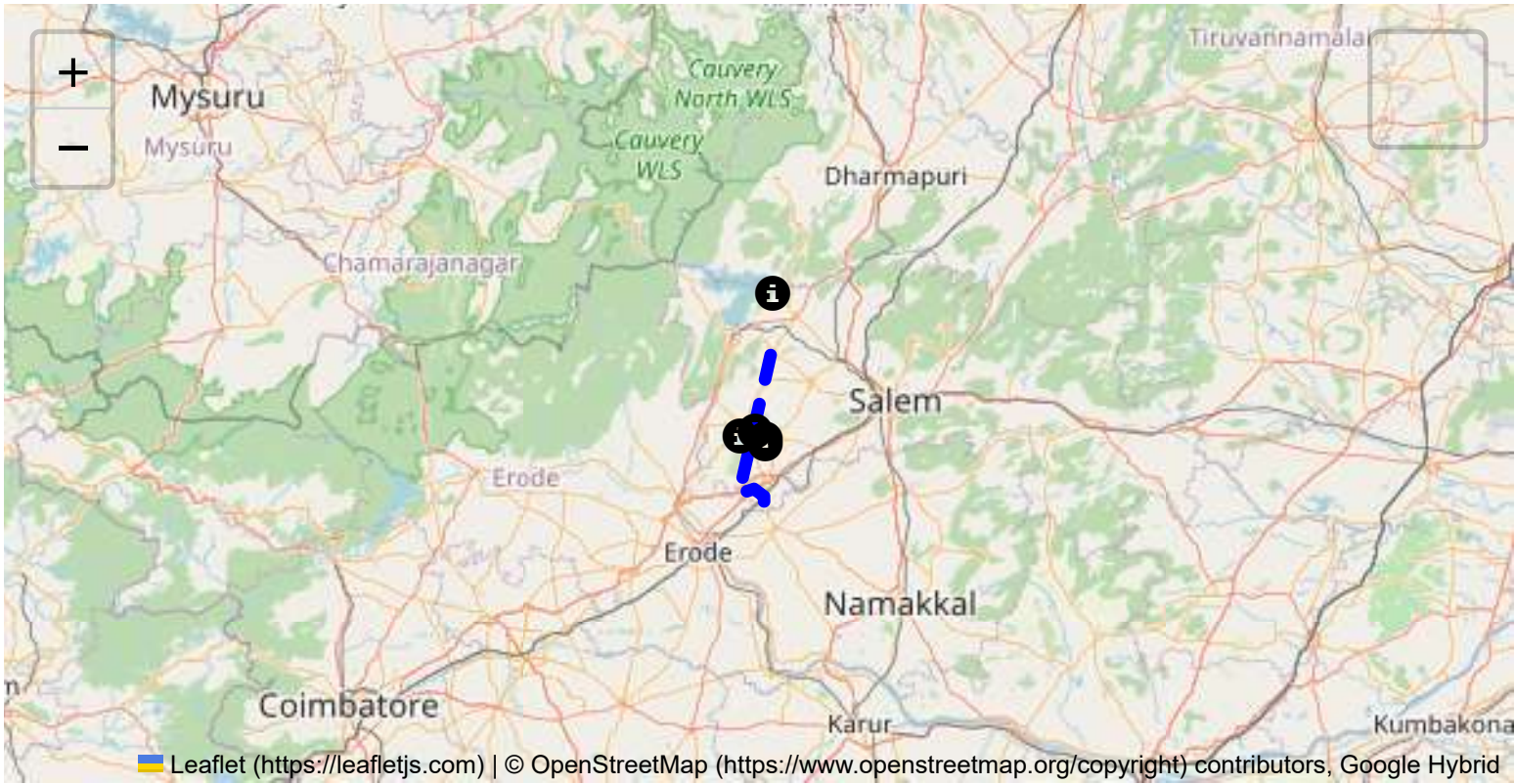
## Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

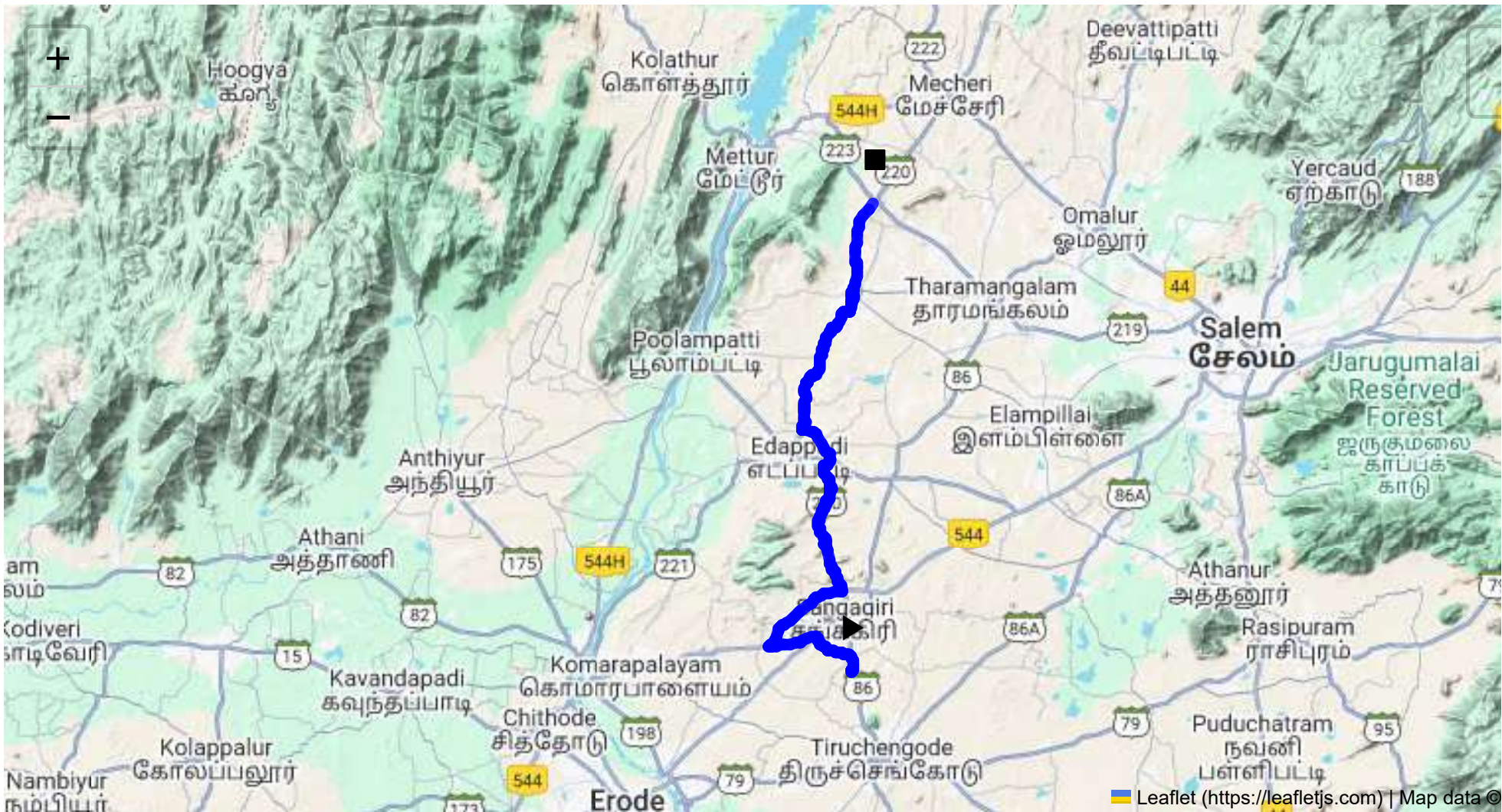
## Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 49.91 km**  
**Estimated Duration: 1.2 hours**  
**Adjusted Duration (Heavy Vehicle): 1.5 hours**  
**Start: (11.4381, 77.8734)**  
**End: (11.758964, 77.889548)**



## Welcome to the Journey Risk Management Study

### 1. Overview of the Route Map

The route runs from Sangagiri to Nangavalli, covering key checkpoints in Sankari RS, Sankari West, and Padaiveedu, spanning approximately 49.91 kilometers. It is primarily on national and state highways, designed to accommodate heavy traffic, including trucks.



## 2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Seasonal Weather:** The region experiences a tropical climate with hot summers (March to June), significant monsoon rains (July to September), and mild winters (December to February).
- **Weather-Related Hazards:** During the monsoon season, heavy rainfall can lead to waterlogging and reduced visibility, potentially causing delays. Flash floods and slippery roads are concerns, particularly in low-lying areas.

## 3. Analysis of Traffic Patterns

- **Peak Hours:** Typical congestion during morning (8:00 - 10:00 AM) and evening hours (5:00 - 8:00 PM), particularly near market areas and urban sections.
- **Congestion-Prone Areas:** Sankari RS and Sankari West see tight traffic due to local commerce and commuter traffic. Bhavani Main Road is a major route subject to delays as it serves as a primary conduit for goods and local traffic.

## 4. Assessment of Road Quality and Infrastructure

- **Road Quality:** Paved roads with some sections undergoing frequent maintenance due to the heavy use by trucks and buses. Potholes and uneven surfaces might be present, especially post-monsoon.
- **Infrastructure:** Adequate signage and markings, but roadside amenities are sparse.

## 5. Suggestions for Alternative Routes for Emergencies

- **Alternative Route:** If the main route is closed or blocked, consider using State Highway 161 and 188 as detours. These have less congestion and alternate access to major intersections, though they may add travel time.

## 6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Transport Restrictions:** Specific permits are required for transporting hazardous materials. Compliance with state regulations on timings (typically restricted during peak hours) and safety protocols is mandatory.
- **Weight and Speed Limits:** Adherence to statutory weight limits and speed regulations (max 60 km/h for trucks) is critical to avoid fines and ensure safety.

## 7. Overview of Historical Incidents

- **Incident Reports:** Historical data suggests occasional accidents, mainly due to speeding and poor weather conditions. No major hazardous material spill incidents have been recorded recently, but vigilance is advised.

## 8. Environmental Considerations and Sensitive Areas

- Environmental Zones:** The route does not pass through protected environmental zones. However, areas near agricultural fields require caution to prevent contamination risks.
- Noise and Air Pollution:** Consideration should be given to minimizing disturbances through inhabited areas.

## 9. Analysis of Communication Coverage

- Mobile Signals:** The route generally has good mobile network coverage, but potential dead zones may exist in rural stretches, particularly between Sankari RS and Padaiveedu.

## 10. Estimated Emergency Response Times

- Emergency Services:** Response times may range from 20 to 40 minutes in rural areas and are quicker in urban sections due to proximity to local police and medical facilities. Nearest major hospital is located in Erode, approximately 30 km away from the route midpoint.

## 11. Overall Summary of Risk Assessment

The Sangagiri to Nangavalli route poses moderate risks common to heavy vehicle travel in rural India, such as weather-related hazards and congestion. Regular maintenance issues and the presence of small urban areas add complexity. However, the route is manageable with careful planning and adherence to regulations. In case of emergencies, viable alternatives exist, though they may extend travel time. Incorporating real-time updates on traffic and weather, along with strict compliance to local transport regulations, will serve to minimize potential risks inherent in the transport of hazardous materials.

### Risk Assessment - Turns

|   | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|---|------------|------------|--------------------|-------------|
| 0 | Roundabout | High       | 11.58300, 77.85813 | 15 KM/Hr    |
| 1 | Roundabout | High       | 11.60348, 77.83967 | 15 KM/Hr    |
| 2 | Turn       | Medium     | 11.43822, 77.87348 | 30 KM/Hr    |
| 3 | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 4 | Turn       | High       | 11.44029, 77.87544 | 15 KM/Hr    |
| 5 | Turn       | Medium     | 11.44900, 77.87406 | 30 KM/Hr    |
| 6 | Turn       | Medium     | 11.45357, 77.85789 | 30 KM/Hr    |
| 7 | Turn       | Medium     | 11.45352, 77.85698 | 30 KM/Hr    |
| 8 | Turn       | Medium     | 11.45847, 77.85140 | 30 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 9  | Turn       | Medium     | 11.46048, 77.85036 | 30 KM/Hr    |
| 10 | Turn       | Medium     | 11.46303, 77.84945 | 30 KM/Hr    |
| 11 | Turn       | Medium     | 11.46314, 77.84928 | 30 KM/Hr    |
| 12 | Turn       | High       | 11.45542, 77.81725 | 15 KM/Hr    |
| 13 | Turn       | High       | 11.45634, 77.81673 | 15 KM/Hr    |
| 14 | Blind Spot | Blind Spot | 11.49390, 77.86748 | 10 KM/Hr    |
| 15 | Turn       | Medium     | 11.49404, 77.86724 | 30 KM/Hr    |
| 16 | Turn       | Medium     | 11.50866, 77.86053 | 30 KM/Hr    |
| 17 | Turn       | Medium     | 11.55039, 77.85511 | 30 KM/Hr    |
| 18 | Turn       | Medium     | 11.55460, 77.85624 | 30 KM/Hr    |
| 19 | Turn       | High       | 11.57893, 77.85430 | 15 KM/Hr    |
| 20 | Turn       | Medium     | 11.57903, 77.85441 | 30 KM/Hr    |
| 21 | Turn       | Medium     | 11.58022, 77.85744 | 30 KM/Hr    |
| 22 | Turn       | Medium     | 11.60727, 77.84032 | 30 KM/Hr    |
| 23 | Turn       | Medium     | 11.60972, 77.84164 | 30 KM/Hr    |
| 24 | Turn       | Medium     | 11.64064, 77.84611 | 30 KM/Hr    |
| 25 | Turn       | Medium     | 11.64535, 77.85122 | 30 KM/Hr    |
| 26 | Turn       | Medium     | 11.66467, 77.85599 | 30 KM/Hr    |
| 27 | Turn       | Medium     | 11.67301, 77.86024 | 30 KM/Hr    |
| 28 | Turn       | Medium     | 11.67326, 77.86058 | 30 KM/Hr    |
| 29 | Turn       | Medium     | 11.67373, 77.86230 | 30 KM/Hr    |
| 30 | Turn       | High       | 11.68490, 77.86893 | 15 KM/Hr    |
| 31 | Turn       | High       | 11.68472, 77.87281 | 15 KM/Hr    |
| 32 | Turn       | Medium     | 11.69280, 77.87439 | 30 KM/Hr    |
| 33 | Turn       | High       | 11.69458, 77.87589 | 15 KM/Hr    |
| 34 | Turn       | Medium     | 11.69475, 77.87583 | 30 KM/Hr    |
| 35 | Turn       | High       | 11.69494, 77.87544 | 15 KM/Hr    |
| 36 | Blind Spot | Blind Spot | 11.69510, 77.87550 | 10 KM/Hr    |
| 37 | Turn       | High       | 11.69553, 77.87440 | 15 KM/Hr    |
| 38 | Turn       | High       | 11.69915, 77.87478 | 15 KM/Hr    |
| 39 | Turn       | Medium     | 11.69922, 77.87494 | 30 KM/Hr    |

## Emergency Locations

|   | type     | name                                           | coordinates            | speed_limit | risk_level |
|---|----------|------------------------------------------------|------------------------|-------------|------------|
| 1 | hospital | Parusuraman Hospital                           | 11.697164, 77.877792   | 30 km/h     | Medium     |
| 2 | hospital | JP Nature Cure Hospital Siddha and Naturopathy | 11.7002949, 77.8770897 | 30 km/h     | Medium     |
| 3 | hospital | SSS Hospital                                   | 11.7501471, 77.8832809 | 30 km/h     | Medium     |
| 4 | clinic   | DSR Clinic                                     | 11.7603176, 77.8908853 | 30 km/h     | Medium     |

## Crowded Spots

|   | type   | name                       | coordinates            | speed_limit | risk_level |
|---|--------|----------------------------|------------------------|-------------|------------|
| 0 | school | KRP Matric. Hr. Sec School | 11.4546193, 77.8142445 | 30 km/h     | Medium     |

## Route Photos of Risky Spots



**Risk Type:** Roundabout

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.58300, 77.85813





**Risk Type:** Roundabout  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.60348, 77.83967



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.44029, 77.87544





**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.44900, 77.87406**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.45357, 77.85789**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.45352, 77.85698**





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45847, 77.85140



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.46048, 77.85036





**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.46303, 77.84945**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.46314, 77.84928**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.45542, 77.81725**





Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45634, 77.81673



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.49390, 77.86748





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.49404, 77.86724



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.50866, 77.86053



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.55039, 77.85511





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.55460, 77.85624



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.57893, 77.85430





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.57903, 77.85441



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.58022, 77.85744



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.60727, 77.84032





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.60972, 77.84164



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.64064, 77.84611





**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.64535, 77.85122**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.66467, 77.85599**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.67301, 77.86024**





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.67326, 77.86058



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.67373, 77.86230





**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.68490, 77.86893**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.68472, 77.87281**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.69280, 77.87439**





Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.69458, 77.87589



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69475, 77.87583





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.69494, 77.87544



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.69510, 77.87550



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.69553, 77.87440





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.69915, 77.87478



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.69922, 77.87494



